

Elasticsearch API basics

```
PUT my_index #create an index
GET _cat/indices #list all indices

POST my_index/_doc/ #index & store document
{
  "title": "Monty Python and the Holy Grail",
  "description": "King Arthur and his Knights
encounter many, very silly obstacles."
}

PUT my_index/_doc/1 #index with provided ID
{
  "title": "The Meaning of Life",
  "description": "The team takes a look at
stages of life in a uniquely silly way."
}
GET my_index #get index details
GET my_index/_count #count docs in index
```

```
PUT my_index/_create/42 #safeguard for
overriding
{
  "title": "Life of Brian",
  "description": "..."
}

POST my_index/_update/1 #update fields
{
  "doc": {
    "description": "the funniest movie!"
  }
}

GET my_index/_doc/1 #get by ID

DELETE my_index/_doc/1 #doc deletion
DELETE my_index #index deletion
```

Index settings

```
PUT my_index
```

```
{
  "settings": { # index creation settings
    "number_of_shards": 4, # the number of shards
    "number_of_replicas": 2, # The number of replicas per shard
    "index.max_result_window": 20000, # max number of results to search for
    "index" : {
      "similarity" : { # query-doc similarity function (variation of cosine TF-IDF)
        "default" : {
          "type" : "BM25", "b": 0.5, "k1": 0
        }
      }
    },
    "analysis": { # text preprocessing
      "analyzer": {# the analyzer is the module that preprocesses text
        "std_english": {"type": "standard", "stopwords": "_english_" }
      }
    }
  }
}
```

Bulk indexing

```
PUT my_index/_settings #update index
{
  "index" : {
    "number_of_replicas" : 2, #number of replicas cab be changed dynamically
    "refresh_interval" : -1 #disable frequent refresh, for efficient bulk insertion
    #"refresh_interval" : "2s" #set refresh interval to 2 seconds
  }
}
```

```
POST _bulk #index records in bulk
{ "index" : { "_index" : "my_index", "_id" : "1"}} #metadata
{"title": "Life of Brian", "year": "1979"} #document
{ "index" : { "_index" : "my_index", "_id" : "2"}} #metadata
{"title": "Brazil", "year": "1985"} #document..
```

<https://www.elastic.co/guide/en/elasticsearch/reference/current/indices-update-settings.html>
<https://www.elastic.co/guide/en/elasticsearch/reference/current/docs-bulk.html>

Mappings

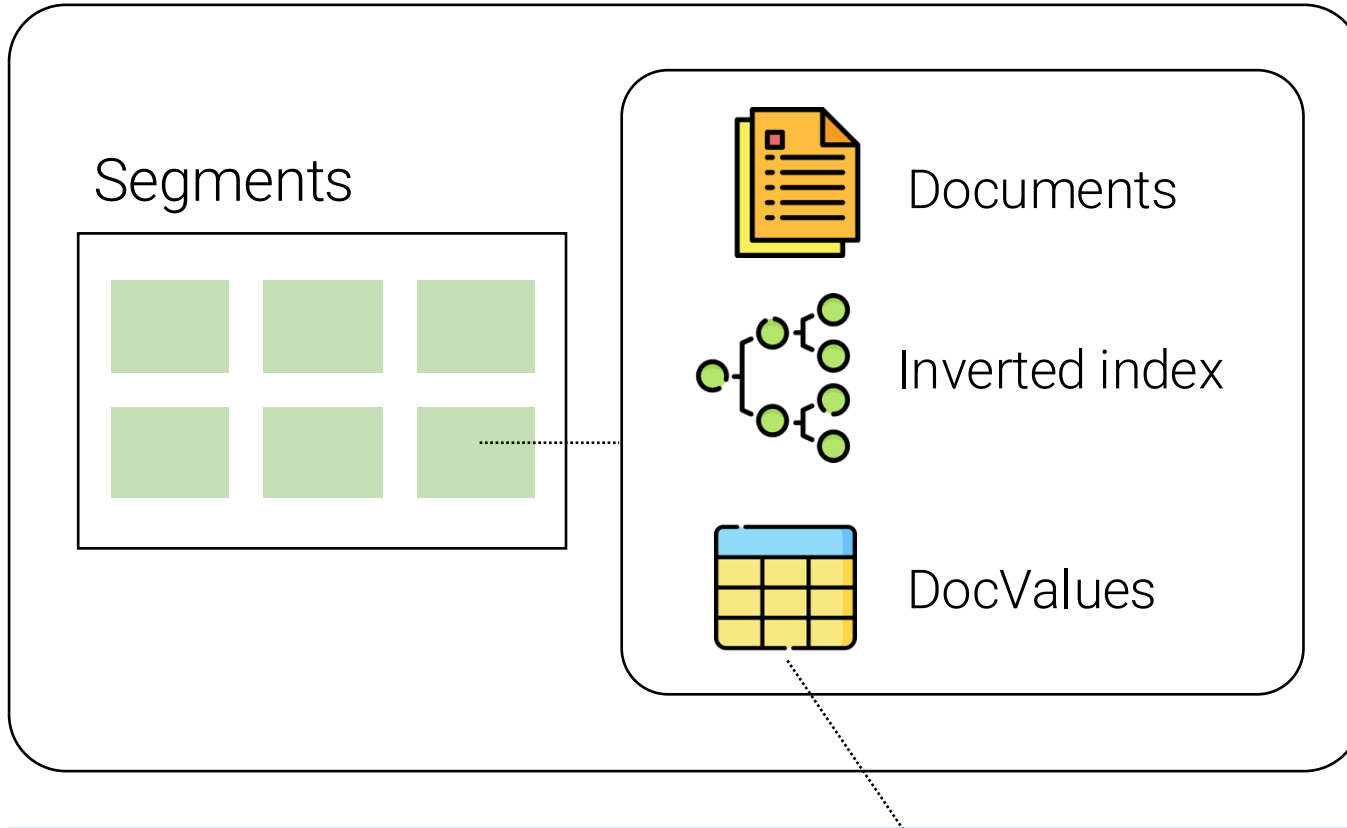
GET my_index/_mapping *#get mapping details*

```
{
  "my_index" : {
    "aliases" : { },
    "mappings" : {
      "properties" : {
        "title" : {
          "type" : "text",
          "fields" : {
            "keyword" : {
              "type" : "keyword",
              "ignore_above" : 256
            }
          }
        }
      }
    }, ...
  }
}
```

- Mappings define how a document and its fields are indexed and stored
- By default, mapping is dynamic (automatically inferred)
- Mappings are defined at index creation (to change a mapping, you need to create a new index, see: **POST _reindex**)

DocValues

ES Shard



- DocValues are “uninverted” indexes
- Tabular data structures that map document IDs to their full fields (either in row or column format)
- Useful for sorting, grouping, faceting, exact search, ...
- Take up more space

```
docID1 : {attr1:val1, attr2:val2, ...} #stored-fields format  
docID2 : {attr1:val3, attr2:val4, ...}
```

```
attr1: {docID1:val1, docID2:val3} #DocValue format  
attr2: {docID1:val2, docID2:val4} ...
```

Mappings

```
GET my_index/_mapping #get mapping details
```

```
{
  "my_index" : {
    "aliases" : { },
    "mappings" : {
      "properties" : {
        "title" : {
          "type" : "text",
          "fields" : {
            "keyword" : {
              "type" : "keyword",
              "ignore_above" : 256
            }
          }
        }
      }
    }, ...
  }
}
```

Create inverted index for this field

Create DocValues for this field

Suggested procedure:

1. Create a test index with no mapping
2. Insert an example document
3. See the dynamic mapping generated automatically
4. Copy the mapping and customize it as needed
5. Delete the test index and create the final index with your custom mapping with **PUT my_index**

Custom mappings

```
"mappings" : {  
  "properties" : {  
    "title" : {  
      "type" : "text",  
      "fields" : {  
        "keyword" : {  
          "type" : "keyword",  
          "ignore_above" : 256  
        }  
      }  
    }  
  }, ...  
}
```

Create Inverted
Index and
DocValues

```
"mappings" : {  
  "properties" : {  
    "title" : {  
      "type" : "text",  
    }, ...  
}
```

Inverted Index

```
"mappings" : {  
  "properties" : {  
    "title" : {  
      "type" : "keyword",  
    }, ...  
}
```

DocValues

```
"mappings" : {  
  "properties" : {  
    "title" : {  
      "enabled": false  
    }, ...  
}
```

Not indexed

Queries

```
#boolean search
GET my_index/_search
{
  "query": {
    "match_phrase": {
      "title": {
        "query": "brian"
      }
    }
  }
}
```

```
#full-text search
GET books/_search
{
  "query": {
    "match": {
      "title": {
        "query": "graal"
      }
    }
  }
}
```

```
#range query
GET books/_search
{
  "range": {
    "year": {
      "gte": 1000,
      "lte": 2000
    }
  }
}
```

<https://www.elastic.co/guide/en/elasticsearch/reference/current/full-text-queries.html>

<https://www.elastic.co/guide/en/elasticsearch/reference/current/query-dsl-match-query.html>

Some mini-tutorials:

<https://github.com/LisaHJung/Part-3-Running-full-text-queries-and-combined-queries-with-Elasticsearch-and-Kibana>

<https://medium.com/elasticsearch/introduction-to-elasticsearch-queries-b5ea254bf455>

Exercise

- Dataset of digitized books

id	author_name	title	country	language	year
14602826	Yearsley, Ann	Poems on several occasions	England	English	1786
14602830	A, T.	A Satyr against Vertue. (A poem: supposed to b...	England	English	1679
14602831	A, T.	The Aeronaut, a poem; founded almost entirely,...	Ireland	English	1816
14602832	Albert, Prince Consort, consort of Victoria, Q...	The Prince Albert, a poem	Ireland	English	1868

Exercise (all from Kibana console!)

Find in the exercise.zip a csv file containing book metadata

1. Create a new index with 3 shards and 2 replicas
2. Insert an example book chosen from the .csv
3. Retrieve the dynamically created mapping
4. Delete the index
5. Create an index with customized mapping
 1. Author name and title should be used for full-text search; country, language and year should be used for filtering (DocValues)
6. In preparation of inserting a bulk of data, disable automatic refreshing
7. Insert 10 books in bulk
8. Get the number of documents that are now in your index. Can you guess the right number?
9. Perform three queries with keyword of your choice:
 1. Full-text
 2. Exact match
 3. Range on years